

Text Summarization

- **Goal:** produce an abridged version of a text that contains information that is important or relevant to a user.
- **Summarization Applications**
 - **outlines or abstracts** of any document, article, etc
 - **summaries** of email threads
 - **action items** from a meeting
 - **simplifying** text by compressing sentences

What to summarize?

Single vs. multiple documents

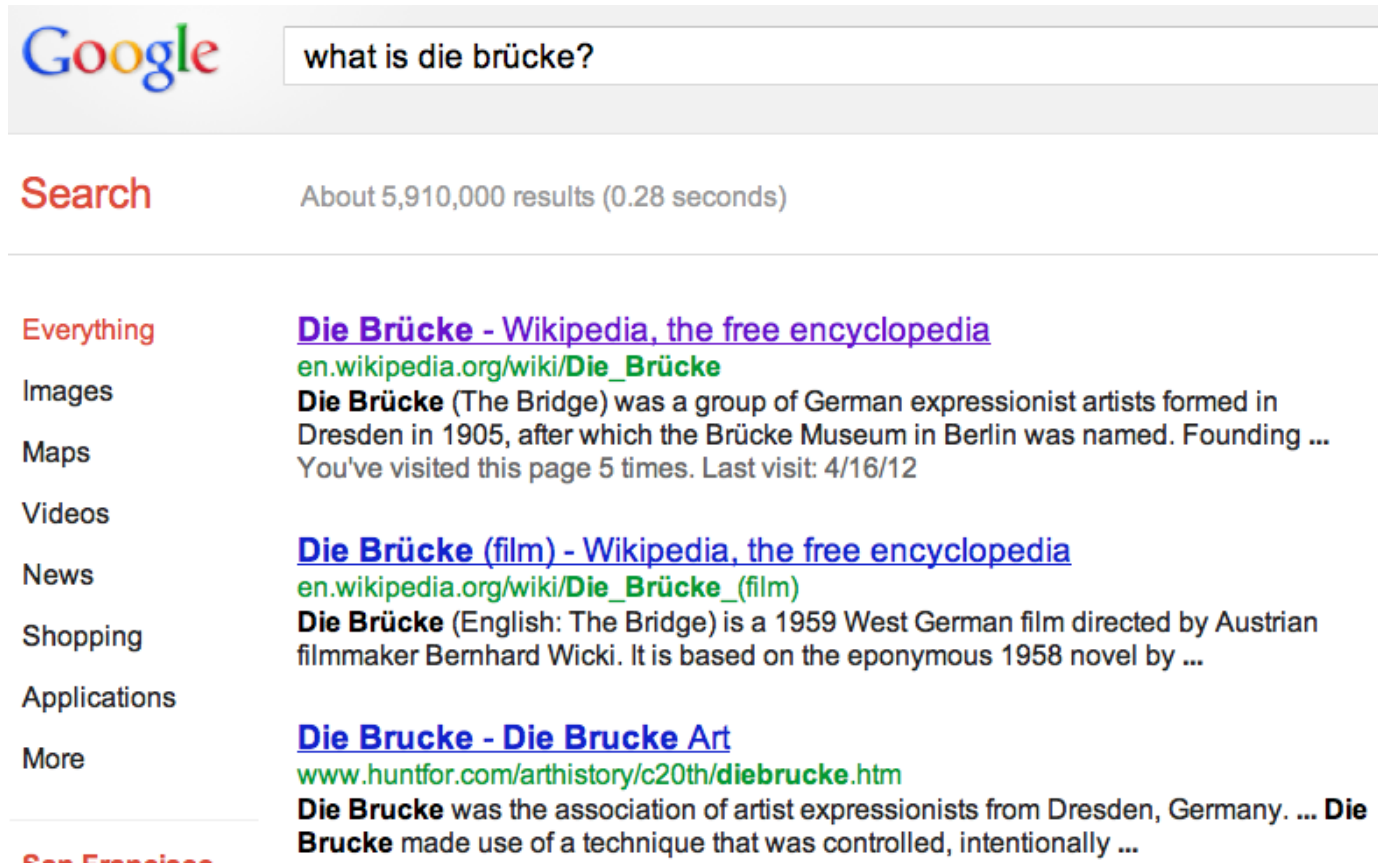
- **Single-document summarization**
 - Given a single document, produce
 - abstract
 - outline
 - headline
- **Multiple-document summarization**
 - Given a group of documents, produce a gist of the content:
 - a series of news stories on the same event
 - a set of web pages about some topic or question

Query-focused Summarization & Generic Summarization

- Generic summarization:
 - Summarize the content of a document
- Query-focused summarization:
 - summarize a document with respect to an information need expressed in a user query.
 - a kind of complex question answering:
 - Answer a question by summarizing a document that has the information to construct the answer

Summarization for Question Answering: Snippets

- Create **snippets** summarizing a web page for a query
 - Google: 156 characters (about 26 words) plus title and link



The screenshot shows a Google search interface. The search bar contains the text "what is die brücke?". Below the search bar, the word "Search" is displayed in red, followed by the text "About 5,910,000 results (0.28 seconds)". On the left side, there is a vertical list of search filters: "Everything", "Images", "Maps", "Videos", "News", "Shopping", "Applications", and "More". The main content area displays three search results. The first result is titled "Die Brücke - Wikipedia, the free encyclopedia" with a green link to "en.wikipedia.org/wiki/Die_Brücke". The second result is titled "Die Brücke (film) - Wikipedia, the free encyclopedia" with a green link to "en.wikipedia.org/wiki/Die_Brücke_(film)". The third result is titled "Die Brücke - Die Brücke Art" with a green link to "www.huntfor.com/arhistory/c20th/diebrücke.htm". Each result is followed by a brief description of the topic.

Google

what is die brücke?

Search About 5,910,000 results (0.28 seconds)

Everything [Die Brücke - Wikipedia, the free encyclopedia](#)
en.wikipedia.org/wiki/Die_Brücke
Die Brücke (The Bridge) was a group of German expressionist artists formed in Dresden in 1905, after which the Brücke Museum in Berlin was named. Founding ...
You've visited this page 5 times. Last visit: 4/16/12

Images

Maps

Videos

News [Die Brücke \(film\) - Wikipedia, the free encyclopedia](#)
[en.wikipedia.org/wiki/Die_Brücke_\(film\)](http://en.wikipedia.org/wiki/Die_Brücke_(film))
Die Brücke (English: The Bridge) is a 1959 West German film directed by Austrian filmmaker Bernhard Wicki. It is based on the eponymous 1958 novel by ...

Shopping

Applications

More [Die Brücke - Die Brücke Art](#)
www.huntfor.com/arhistory/c20th/diebrücke.htm
Die Brücke was the association of artist expressionists from Dresden, Germany. ... **Die Brücke** made use of a technique that was controlled, intentionally ...

San Francisco

Summarization for Question Answering: Multiple documents

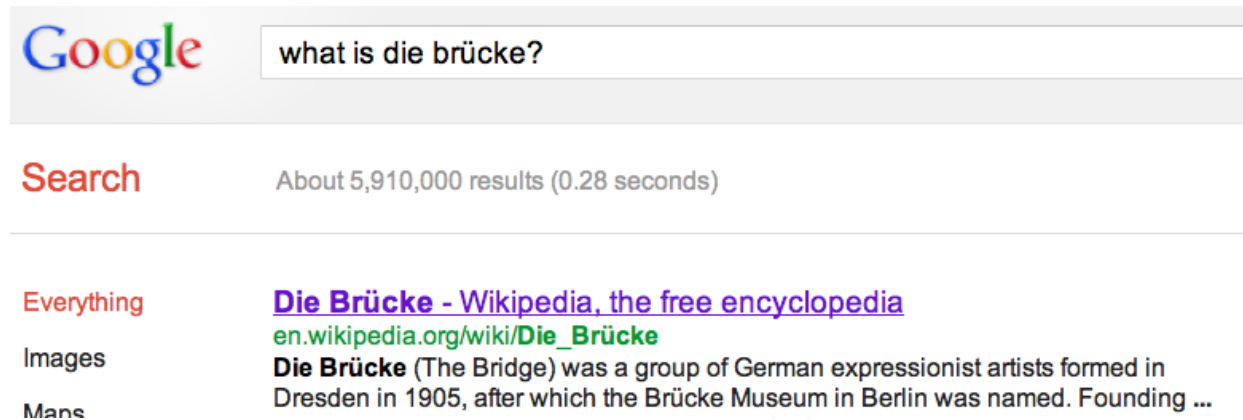
Create **answers** to complex questions summarizing multiple documents.

- Instead of giving a snippet for each document
- Create a cohesive answer that combines information from each document

Extractive summarization & Abstractive summarization

- **Extractive summarization:**
 - create the summary from phrases or sentences in the source document(s)
- **Abstractive summarization:**
 - express the ideas in the source documents using (at least in part) different words

Simple baseline: take the first sentence



Die Brücke

From Wikipedia, the free encyclopedia

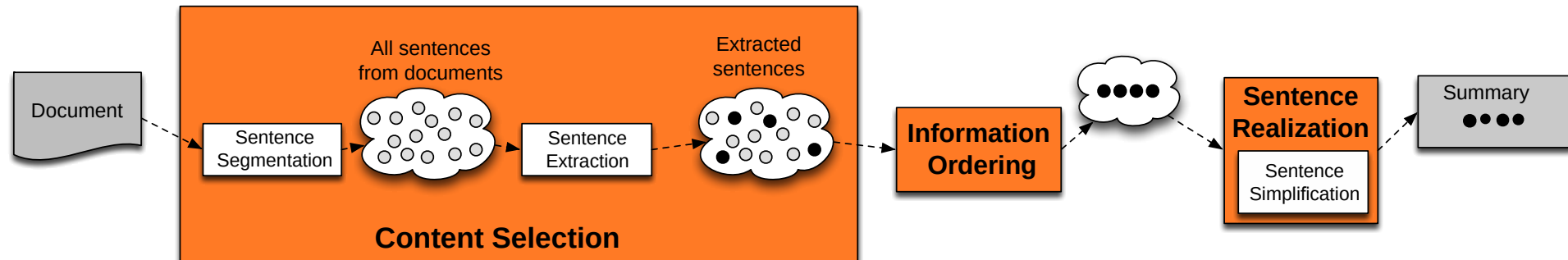
For other uses, see [Die Brücke \(disambiguation\)](#).

Die Brücke (**The Bridge**) was a group of [German expressionist](#) artists formed in [Dresden](#) in 1905, after which the [Brücke Museum in Berlin](#) was named. Founding members were [Fritz Bleyl](#), [Erich Heckel](#), [Ernst Ludwig Kirchner](#) and [Karl Schmidt-Rottluff](#). Later members were [Emil Nolde](#), [Max Pechstein](#) and [Otto Mueller](#). The seminal group had a major impact on the evolution of [modern art](#) in the 20th century and the creation of expressionism.^[1]

Die Brücke is sometimes compared to the [Fauves](#). Both movements shared interests in [primitivist](#) art. Both

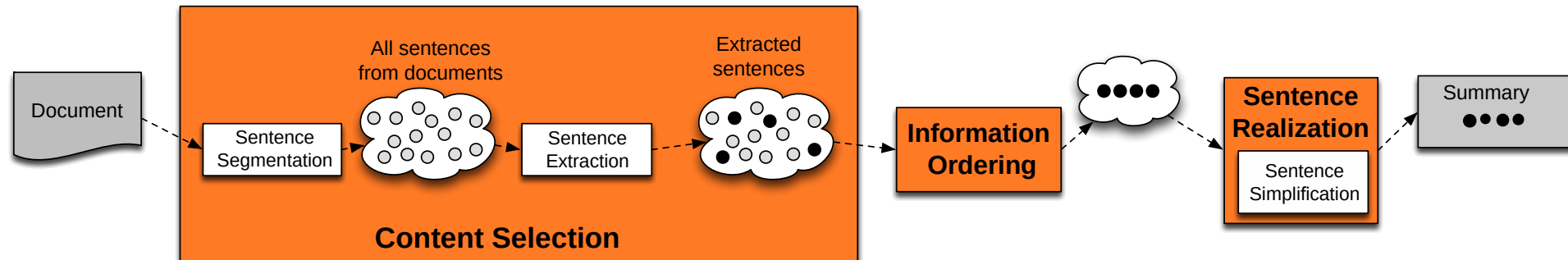
Summarization: Three Stages

1. **content selection**: choose sentences to extract from the document
2. **information ordering**: choose an order to place them in the summary
3. **sentence realization**: clean up the sentences



Basic Summarization Algorithm

1. **content selection**: choose sentences to extract from the document
2. **information ordering**: just use document order
3. **sentence realization**: keep original sentences



Unsupervised content selection

H. P. Luhn. 1958. The Automatic Creation of Literature Abstracts.
IBM Journal of Research and Development. 2:2, 159-165.

- Intuition dating back to Luhn (1958):
 - Choose sentences that have **salient** or **informative** words
- Two approaches to defining salient words
 1. **tf-idf**: weigh each word w_i in document j by tf-idf
$$weight(w_i) = tf_{ij} \cdot idf_i$$
 2. **topic signature**: choose a smaller set of salient words
 - mutual information
 - log-likelihood ratio (LLR) Dunning (1993), Lin and Hovy (2000)

$$weight(w_i) = \begin{cases} 1 & \text{if } -2 \log / (w_i) > 10 \\ 0 & \text{otherwise} \end{cases}$$

Suppose there's a word w appears once in a 30-word document d :

$$C(d) = 1$$

$$N(d) = 30$$

the probability of w in the input = $p(d) = 1/30$

Suppose the background corpus has 4000 words, with w appears 20 times:

$$C(b) = 20$$

$$N(b) = 4000$$

the probability of w in the corpus = $p(b) = 20/4000 = 1/200$

The LLR for a word, generally called $\lambda(w)$, is the ratio between (the probability of observing w in both the input and in the background corpus assuming equal probabilities in both corpora), and the probability of observing w in both assuming different probabilities for w in the input and the background corpus.

The probability of observing w in both the input and in the background corpus assuming equal probabilities in both corpora:

$$[C(d)+C(b)]/[N(d)+N(b)] = p = (1+20)/(30+4000)$$

$$LLR(w) = -2 \times \log \left(\frac{\binom{30}{1} p^1 (1-p)^{29} \times \binom{4000}{20} p^{20} (1-p)^{3980}}{\binom{30}{1} p_d^1 (1-p)^{29} \times \binom{4000}{20} p_b^{20} (1-p)^{3980}} \right)$$

Topic signature-based content selection with queries

Conroy, Schlesinger, and O'Leary 2006

- choose words that are informative either
 - by log-likelihood ratio (LLR)
 - or by appearing in the query

$$weight(w_i) = \begin{cases} 1 & \text{if } -2 \log / (w_i) > 10 \\ 1 & \text{if } w_i \hat{=} question \\ 0 & \text{otherwise} \end{cases} \quad \text{(could learn more complex weights)}$$

- Weigh (Score) a sentence by weight of its words:

$$weight(s) = \frac{1}{|S|} \sum_{w \hat{=} s} weight(w)$$

Supervised content selection

- Given:
 - a labeled training set of good summaries for each document
- Align:
 - the sentences in the document with sentences in the summary
- Extract features
 - position (first sentence?)
 - length of sentence
 - word informativeness, cue phrases
- Train
 - a binary classifier (put sentence in summary? yes or no)
- Problems:
 - hard to get labeled training data
 - alignment difficult
 - performance not better than unsupervised algorithms
- So in practice:
 - **Unsupervised content selection is more common**

ROUGE (Recall Oriented Understudy for Gisting Evaluation)

Lin and Hovy 2003

- Intrinsic metric for automatically evaluating summaries
 - Based on BLEU (a metric used for machine translation)
 - Not as good as human evaluation
 - But much more convenient
- Given a document D, and an automatic summary X:
 1. Have N humans produce a set of reference summaries of D
 2. Run system, giving automatic summary X
 3. What percentage of the bigrams from the reference summaries appear in X?

$$ROUGE - 2 = \frac{\sum_{s \in \{\text{RefSummaries}\}} \sum_{\text{bigrams } i \in S} \min(\text{count}(i, X), \text{count}(i, S))}{\sum_{s \in \{\text{RefSummaries}\}} \sum_{\text{bigrams } i \in S} \text{count}(i, S)}$$

A ROUGE example:

Q: “What is water spinach?”

Human 1: Water spinach is a green leafy vegetable grown in the tropics.

Human 2: Water spinach is a semi-aquatic tropical plant grown as a vegetable.

Human 3: Water spinach is a commonly eaten leaf vegetable of Asia.

- System answer: Water spinach is a leaf vegetable commonly eaten in tropical areas of Asia.

- ROUGE-2 =
$$\frac{3 + 3 + 6}{10 + 9 + 9} = 12/28 = .43$$

Definition questions

Q: What is *water spinach*?

A: Water spinach (*ipomoea aquatica*) is a semi-aquatic leafy green plant with long hollow stems and spear- or heart-shaped leaves, widely grown throughout Asia as a leaf vegetable. The leaves and stems are often eaten stir-fried flavored with salt or in soups. Other common names include *morning glory vegetable*, *kangkong* (Malay), *rau muong* (Viet.), *ong choy* (Cant.), and *kong xin cai* (Mand.). It is not related to spinach, but is closely related to sweet potato and convolvulus.

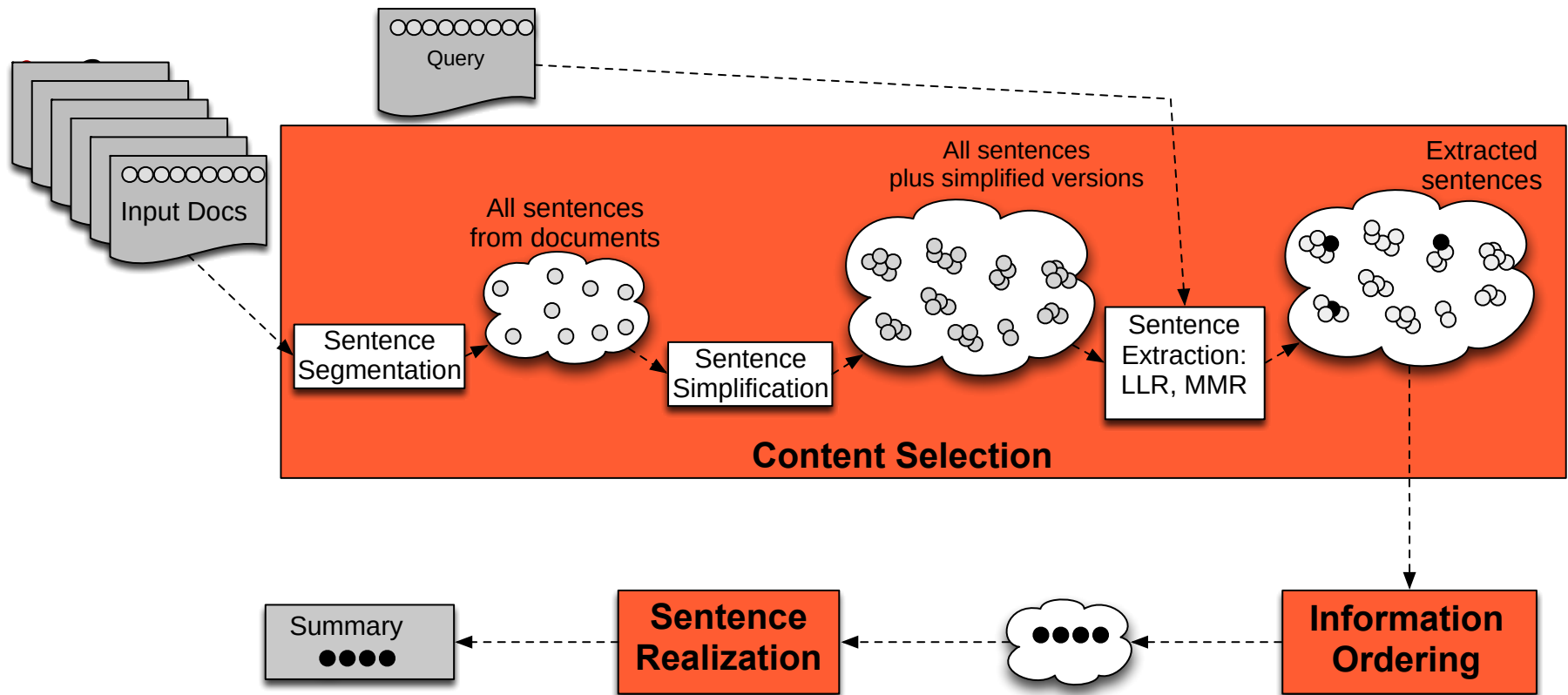
Medical questions

Demner-Fushman and Lin (2007)

Q: In children with an acute febrile illness, what is the efficacy of single medication therapy with acetaminophen or ibuprofen in reducing fever?

A: Ibuprofen provided greater temperature decrement and longer duration of antipyresis than acetaminophen when the two drugs were administered in approximately equal doses.

Query-Focused Multi-Document Summarization



Maximal Marginal Relevance (MMR)

Jaime Carbonell and Jade Goldstein, The Use of MMR, Diversity-based Reranking for Reordering Documents and Producing Summaries, SIGIR-98

- Maximal marginal relevance is an important "diversity based ranking technique", used to maximize the relevance and novelty in finally retrieved top-ranked items. It is highly useful in text-document summarization, answer extraction, document re-ranking, etc.
- An iterative method for content selection from multiple documents

Maximal Marginal Relevance (MMR)

- Iteratively (greedily) choose the best sentence to insert in the summary/answer so far:
 - **Relevant**: Maximally relevant to the user's query
 - high cosine similarity to the query
 - **Novel**: Minimally redundant with the summary/answer so far
 - low cosine similarity to the summary
- Stop when desired length

$$\hat{s}_{MMR} = \max_{s \in D} \lambda \text{sim}(s, Q) - (1-\lambda) \max_{s \in S} \text{sim}(s, S)$$

LLR+MMR: Choosing informative yet non-redundant sentences

- One of many ways to combine the intuitions of LLR and MMR:
 1. Score each sentence based on LLR (including query words)
 2. Include the sentence with highest score in the summary.
 3. Iteratively add into the summary high-scoring sentences that are not redundant with summary so far.

Information Ordering

- Chronological ordering:
 - Order sentences by the date of the document (for summarizing news)..
(Barzilay, Elhadad, and McKeown 2002)
- Coherence:
 - Choose orderings that make neighboring sentences similar (by cosine).
 - Choose orderings in which neighboring sentences discuss the same entity
(Barzilay and Lapata 2007)

Domain-specific answering: The Information Extraction method

- a good **biography** of a person contains:
 - a person's **birth/death, fame factor, education, nationality** and so on
- a good **definition** contains:
 - **genus** or **hypernym**
 - *The Hajj is a type of ritual*
- a **medical answer about a drug's use** contains:
 - **the problem** (the medical condition),
 - **the intervention** (the drug or procedure), and
 - the **outcome** (the result of the study).

Information that should be in the answer for 3 kinds of questions

Definition	
genus	The Hajj is a type of ritual
species	the annual hajj begins in the twelfth month of the Islamic year
synonym	The Hajj, or Pilgrimage to Mecca, is the central duty of Islam
subtype	Qiran, Tamattu', and Ifrad are three different types of Hajj
Biography	
dates	was assassinated on April 4, 1968
nationality	was born in Atlanta, Georgia
education	entered Boston University as a doctoral student
Drug efficacy	
population	37 otherwise healthy children aged 2 to 12 years
problem	acute, intercurrent, febrile illness
intervention	acetaminophen (10 mg/kg)
outcome	ibuprofen provided greater temperature decrement and longer duration of antipyresis than acetaminophen when the two drugs were administered in approximately equal doses